GlucoTwin AI

A Personalized Digital Twin for Type 2 Diabetes

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Problem Statement & Healthcare Use Case

Problem: Managing Type 2 Diabetes requires constant monitoring. Unpredicted glucose spikes can lead to severe long-term complications.

Use Case: GlucoTwin AI acts as a patient-specific Digital Twin that fuses clinical health records with time-series Continuous Glucose Monitor (CGM) and lifestyle data. It utilizes machine learning to predict potential glucose spikes up to two hours in advance, allowing for proactive, preventative action.

Technical Stack & AI/ML Models

- Frontend: Streamlit, Plotly (for interactive Radar & Time-Series charts)
- Backend: Python, Pandas, NumPy
- AI/ML Framework: Scikit-learn (Random Forest Classifier) for time-series forecasting and spike prediction.
- Reporting: fpdf2 and kaleido for automated, offline PDF clinical report generation.

Our model extracts time-series features (like rolling averages and lagged glucose changes) and calculates the exact probability of a dangerous glucose spike.

System Architecture

1. Frontend (Streamlit): The user inputs medical vitals and uses What-If sliders to simulate lifestyle changes.

2. Backend Engine: Generates a mathematically synthetic 24-hour CGM curve aligned with the user's clinical inputs.

3. Machine Learning Core: Feeds time-series features into a trained Random Forest Classifier.

4. Offline Reporting Engine: Captures headless screenshots of interactive Plotly graphs to embed into a structured FPDF-generated clinical report.

Project Details and Outcomes

Outcomes:

- Successfully built an interactive, end-to-end digital twin.
- Implemented physiological What-If simulations for Carbs, Activity, and Renal Filtration (Hydration).
- Developed an automated PDF reporting system for physicians.
- Deployed live via Streamlit Community Cloud for public accessibility.

GlucoTwin AI demonstrates the immense potential of AI in proactive metabolic health management.